Variance-Corrected Multi-Asset Equity Simulation with Hybrid Hidden Markov Marginals

Abstract

Synthetic multi-asset equity data must reproduce each asset's return distribution and its relationship with the market. A common approach adds a synthetic full-return path to a market factor as though the synthetic path were a residual. Because that path already contains the asset's total variance, the construction adds market variance a second time. We derived a correction that centers and rescales each synthetic path before adding the market factor. We tested the method on 423 non-market assets in a 424-asset United States equity and exchange-traded-fund universe. The correction preserved each generator's target variance, recovered its calibrated market loading with low error, and retained heavy-tailed return distributions without fitting a second model to the regression residuals. On 416 complete asset histories held out from 2025, it increased the mean Kolmogorov–Smirnov pass rate from 83.5% to 87.0% and reduced the median ratio of synthetic to observed variance from 1.29 to 0.97. Its one-day left-tail 99% Value-at-Risk exceedance rate was similar to that of the best residual-fit method, although both exceeded the nominal rate. We generated the per-asset paths with a hidden Markov model whose jump-duration variant lengthened visits to extreme-return states. On a decade of daily data for a broad-market exchange-traded fund (ticker SPY), this variant improved the match to volatility clustering relative to the no-jump model while maintaining strong distributional fit. We selected it as a practical compromise, not because it outperformed every baseline. The multi-asset method still omits residual dependence between assets, which can overstate the benefit of daily rebalancing. Code and fitted per-asset models are released with the paper.

Keywords: hidden Markov model, jump-duration mechanism, single-index model, synthetic financial data, variance correction, Value-at-Risk coverage check, volatility clustering, heavy-tailed marginals

1. Introduction

Researchers and practitioners often use synthetic paths to stress-test risk models against unobserved market scenarios, validate portfolio optimization algorithms, and evaluate allocation strategies across many simulations (Assefa et al., 2020; Jordon et al., 2022). However, synthetic financial data often fail to capture important statistical features of real returns (Stenger et al., 2024). Daily excess growth rates for a broad-market exchange-traded fund (ticker SPY) from 2014–2024 illustrate three widely documented stylized facts (Mandelbrot, 1963;

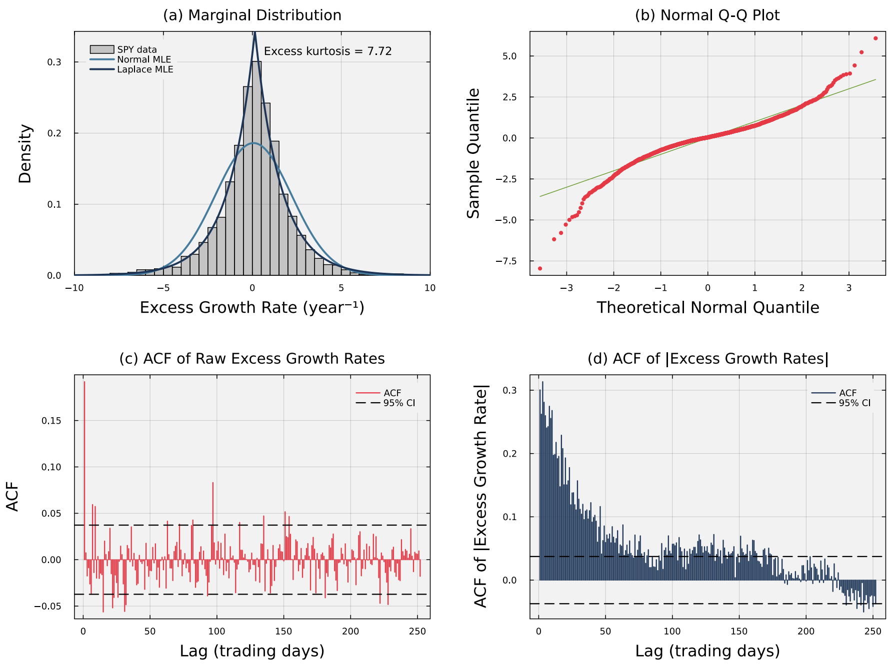

Figure 1: **Empirical stylized facts of SPY daily excess growth rates (2014–2024).** Panel (a) shows the heavy-tailed return density; the Laplace maximum-likelihood fit tracks the empirical peak and tails while the Gaussian fit does not. Panel (b) shows departure from the diagonal in a normal quantile-quantile plot at both extremes. Panel (c) shows the autocorrelation function of raw excess growth rates. Individual correlations are small, although the Ljung–Box test detects joint serial dependence. Panel (d) shows the autocorrelation function of absolute excess growth rates, which decays slowly across the reported lags.

Cont, 2001) (Fig. 1). Most observed returns lie close to zero, but extreme returns occur much more often than a Gaussian model predicts (Fig. 1a and Fig. 1b). Raw growth rates show little correlation across days (Fig. 1c). Absolute growth rates, in contrast, remain positively correlated over longer periods (Fig. 1d), so large and small changes cluster over time. Matching the marginal return distribution alone therefore does not reproduce how volatility evolves over time. Multi-asset simulations must also reproduce cross-asset dependence to evaluate portfolio risk and performance under different rebalancing strategies (Embrechts et al., 2002; Bouchey et al., 2015). Building a multi-asset simulation therefore starts with choosing a single-asset generator that reproduces each asset's return distribution and volatility dynamics.

Single-asset generators differ in the return features they reproduce and in the effort required to fit them across a large asset universe. Generalized autoregressive conditional heteroskedasticity (GARCH) models capture volatility clustering, but may require heavy-tailed innovations to reproduce extreme returns (Engle, 1982; Bollerslev, 1986). Stochastic-volatility models represent persistent changes in volatility, while jump-diffusion models represent abrupt extremes (Heston, 1993; Merton, 1976; Kou, 2002). Jumps alone do not produce persistent volatility. Deep generative models can learn flexible return distributions and temporal patterns. Some generative adversarial networks (GANs) reproduce volatility clustering and other path-dependent features (Wiese et al., 2020; Ni et al., 2021). Other evaluations of GANs and the time-series generative adversarial network (TimeGAN) miss the absolute-return autocorrelation associated with volatility clustering (Takahashi et al., 2019; Yoon et al., 2019; Kwon and Lee, 2024). Hidden Markov models provide interpretable market regimes, but standard transition rules can leave high-volatility states too quickly to reproduce long clusters of large returns (Rabiner and Juang, 1986; Rydén et al., 1998; Bulla and Bulla, 2006). Choosing a single-asset model requires balancing distributional fit, volatility persistence, interpretability, and the effort required to fit the model across hundreds of assets. Portfolio simulation adds a separate requirement because independently fitted single-asset models do not specify how assets move together.

Multivariate hidden-state models allow assets to share market regimes, but become harder to fit as the number of assets grows (Dias et al., 2010). Copula models combine separately fitted asset distributions with a model of cross-asset dependence and can capture nonlinear and tail relationships (Embrechts et al., 2002; Cherubini et al., 2004; Aas et al., 2009). However, copula models require choosing a dependence structure, and standard static copulas do not describe how dependence changes across days. Factor models are easier to apply to large asset universes because each asset is linked to a small number of shared factors instead of modeling every relationship between pairs of assets (Sharpe, 1963; Ross, 1976; Fama and French, 1993). The single-index model separates each asset's return into a market component and an asset-specific residual. After fitting the model, the historical residuals are easy to calculate. The residuals can be resampled or used to fit a separate generator for each asset. Fitting residual generators is a direct solution, but adds a second fitting step for every asset. Generating residual paths independently also misses sector, style, and other relationships not explained by the market. Reusing an existing full-return generator avoids the second fit and preserves the fitted return distribution and temporal behavior. However, a full-return draw already contains the asset mean and market-related variance. Treating the full-return draw as a residual can shift the mean and count market variance twice. The problem is not how to calculate residuals. The problem is whether an existing full-return generator can be reused without shifting the mean or inflating the variance. Dependence among the residuals remains a separate problem.

In this study, we explored how to build a multi-asset generator from separately fitted single-asset components and a shared market path without shifting expected returns or overcounting market variance. We used a hidden Markov model (HMM) for each asset, with Laplace-quantile states, Student-t emissions, and counted transitions. Temporary episodes in the extreme states distinguished the hidden Markov model with jumps (HMM-WJ) from the hidden Markov model with no jumps (HMM-NJ). On a decade of SPY data, HMM-WJ reduced absolute-return autocorrelation error from 0.059 to 0.053 relative to HMM-NJ while retaining a 98.3% in-sample Kolmogorov–Smirnov pass rate. GARCH achieved a lower autocorrelation error of 0.031 but did not match the return distribution, so HMM-WJ served as a practical per-asset generator rather than a uniformly superior model. For multi-asset composition, we centered and rescaled each asset-level path before adding a shared SPY path. The correction removed the mean shift and prevented the market contribution to variance from being counted twice without fitting another model to each asset's regression residuals. We tested the correction on 423 non-market assets in a 424-asset universe and set the jump probability to zero to isolate the composition rule. With all 2014–2024 parameters held fixed, the correction increased the mean Kolmogorov–Smirnov pass rate from 83.5% to 87.0% across 416 complete asset histories from 2025 and reduced the median ratio of synthetic to observed variance from 1.29 to 0.97. Independent residual generation did not reproduce residual dependence and could overstate returns from daily rebalancing.

2. Related Work

Single-asset return models treat changing volatility, market regimes, and extreme moves in different ways. Autoregressive conditional heteroskedasticity (ARCH) and generalized autoregressive conditional heteroskedasticity (GARCH) models allow conditional variance to change over time (Engle, 1982; Bollerslev, 1986). Even with Gaussian innovations, the changing variance can produce an unconditional return distribution with excess kurtosis. Heavy-tailed innovations can improve the tail fit when Gaussian GARCH understates the frequency of extreme returns. Stochastic-volatility models make volatility a latent process (Stein and Stein, 1991; Heston, 1993; Kim et al., 1998). Jump-diffusion models add discontinuous price moves (Merton, 1976; Kou, 2002), but a jump component alone does not explain persistent volatility. Markov-switching ARCH and GARCH models combine discrete regimes with changing conditional variance (Hamilton and Susmel, 1994; Klaassen, 2002). Hidden-state models represent returns with a small number of market regimes. In a hidden Markov model (HMM), each latent state has a separate return distribution. Gaussian-emission HMMs can reproduce heavy tails and weak raw-return autocorrelation, but may still miss volatility clustering (Hamilton, 1989; Rydén et al., 1998). Hidden semi-Markov models allow more flexible state durations and can improve volatility persistence (Bulla and Bulla, 2006). An HMM does not create a jump process by itself. Rare extreme moves must come from the state-dependent return distributions or from a transition rule designed for such events. Deep generative models learn return distributions and temporal features without specifying a small set of market regimes or a parametric return process. Generative adversarial networks (GANs) are one class of deep generative model. Quant GANs have reproduced several financial time-series features, including volatility clustering and leverage effects (Wiese et al., 2020). Sig-Wasserstein GANs use path signatures to learn features that depend on the order of observations (Ni et al., 2021). In contrast, evaluations of GAN paths, including paths from the time-series generative adversarial network (TimeGAN), have found too little autocorrelation in absolute returns (Takahashi et al., 2019; Yoon et al., 2019; Kwon and Lee, 2024). The conflicting results show that flexible architectures do not guarantee a match to financial stylized facts. Model design, training, and evaluation all matter (Assefa et al., 2020; Stenger et al., 2024).

Multi-asset generation also requires a model of cross-asset dependence. Multivariate GARCH models estimate time-varying covariance matrices. The dynamic conditional correlation model reduces the number of parameters by separating individual volatility models from correlation dynamics (Engle, 2002; Bauwens et al., 2006). Copulas separate each asset's return distribution from the dependence model. Dynamic copulas allow cross-asset dependence to change over time, while vine copulas provide flexible high-dimensional constructions (Patton, 2006; Aas et al., 2009). Factor models reduce dimension by linking each asset to a small number of shared factors (Sharpe, 1963; Ross, 1976; Fama and French, 1993). Shared-state HMMs and multivariate GANs can also generate joint market moves (Dias et al., 2010; Rizzato et al., 2023). A standard factor model can resample historical residuals or fit a separate residual generator for every asset. This paper addresses a narrower problem. We start with single-asset generators already fitted to full returns and ask whether they can be reused in a factor model. The regression residuals are easy to calculate. The difficulty is that a full-return draw already contains the asset mean and market-related variance. Using that draw as a residual can shift the mean and count part of the variance twice. The correction developed here removes the mean shift and variance double count without fitting a second generator for every asset. It does not model dependence among the remaining residuals.

3. Methods

We fitted a marginal generator for each asset and then combined the generated paths with a variance-corrected single-index model. The dataset contained 424 United States-listed equities and exchange-traded funds observed from 2014 through 2024. The single-asset analysis focused on the broad-market exchange-traded fund with ticker SPY. It used 2,767 daily volume-weighted average prices for fitting, which yielded 2,766 growth rates, and the next 250 prices from 2025, which yielded 249 holdout growth rates. The multi-asset analysis used daily closing prices. Of the 423 non-market assets in the training universe, 416 had complete 2025 histories and were included in the multi-asset holdout evaluation. We defined the excess growth rate $G_{i,j}$ for ticker i between trading days $j-1$ and j from prices $P_{i,j-1}$ and $P_{i,j}$.

$$G_{i,j} \;\equiv\; \left(\frac{1}{\Delta t}\right) \ln\!\left(\frac{P_{i,j}}{P_{i,j-1}}\right) \;-\; r_f. \tag{1}$$

Here, $\Delta t = 1/252$ is the fraction of a trading year spanned by one trading day, and r_f is a constant continuously compounded risk-free rate. We used

$r_f = 0.043$ for the SPY training period and $r_f = 0.0421$ for its 2025 holdout, based on Treasury Separate Trading of Registered Interest and Principal of Securities (STRIPS) bond yields. The excess growth rate has units of year^{-1} and is time-additive under continuous compounding.

3.1. Per-asset hidden Markov marginal generator

For each asset, the hidden Markov model with jumps (HMM-WJ) converted the continuous return series into a sequence of N ordered return regimes. At each time step, the model first selected a regime and then drew a return from that regime's distribution (Fig. 2). We represented the model as $\mathcal{M} = (\mathcal{S}, \mathcal{O}, \mathbf{T}, \mathbf{E}, \bar{\pi})$ (Rabiner and Juang, 1986). The hidden state $S_t \in \mathcal{S} = \{1, \ldots, N\}$ identifies the regime at time t, and the observed value is the excess growth rate G_t from Eq. (1). We omit the ticker index i when discussing one asset. The transition matrix $\mathbf{T}$ sets the probabilities of moving between regimes, $\mathbf{E}$ gives the return distribution within each regime, and $\bar{\pi}$ gives the initial regime probabilities. We ordered the regimes from the most negative to the most positive returns. Their boundaries were equally spaced quantiles of a Laplace distribution fitted to the asset's in-sample returns. Specifically, $Q_k = F_L^{-1}(k/N;\, \mu_L, b_L)$, where μ_L and b_L are the fitted location and scale. We set $Q_0 = -\infty$ and $Q_N = +\infty$ and assigned return G_t to state k when $Q_{k-1} < G_t \leq Q_k$. The Laplace distribution captured the sharp concentration of small price movements in Fig. 1a (Kotz et al., 2001; Toth and Jones, 2019). We computed its closed-form quantiles for all 424 tickers (Bilmes, 1998).

The within-state emission followed a location-scale Student-t distribution.

$$G_t \mid S_t = k \;\sim\; \mu_k + \sigma_k \cdot t_\nu, \qquad \nu = 5, \tag{2}$$

where t_ν denotes a standard Student-t random variable with ν degrees of freedom. For observations assigned to state k, μ_k is the sample mean and σ_k is the sample standard deviation. A standard t_ν has variance $\nu/(\nu - 2)$, so σ_k is a scale parameter rather than the emission standard deviation. At $\nu = 5$, the emission standard deviation is $\sigma_k\sqrt{5/3}$. We selected $\nu = 5$ from a sensitivity sweep. We used $\nu = \infty$ for the Normal baseline; Table 1 gives the explored range. At $\nu = 5$, simulated kurtosis was 7.5 against the observed 7.7, compared with 5.5 under Normal emissions. The corresponding Student-t Kolmogorov–Smirnov (KS) and Anderson–Darling (AD) pass rates were 98.3% and 94.8%. We also compared HMM-WJ with a hidden semi-Markov model (HSMM) baseline adapted from Bulla and Bulla (2006) (Supplementary Table S3; Sec. S5). The emission and HSMM comparisons separated the effects of heavy-tailed emissions and the jump-duration mechanism. We estimated the transition matrix $\mathbf{T}$ by counting observed transitions, which avoided the initialization sensitivity of iterative expectation-maximization (Bilmes, 1998). The counted matrix was concentrated near its diagonal, but individual states, including tail states, usually lasted only 1–2 days (Supplementary Fig. S5 in Online Appendix S4). The counted transition probabilities caused the model to leave extreme-return states too quickly to reproduce the observed volatility clustering (Bulla and Bulla, 2006).

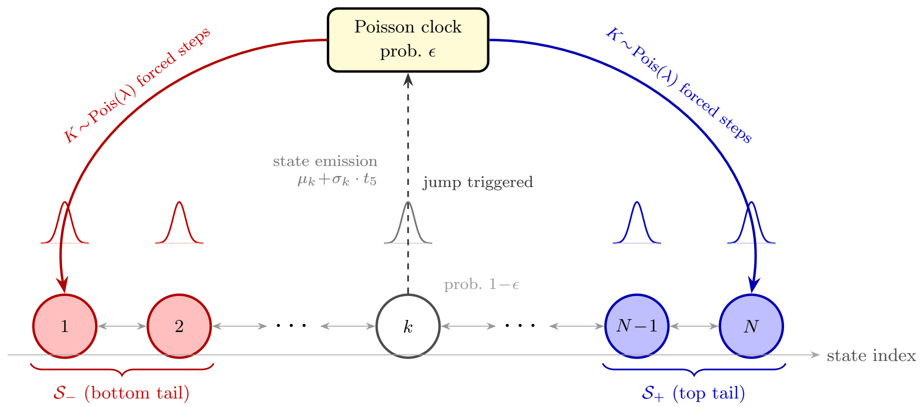

Figure 2: **Architecture of the hidden Markov model with jumps (HMM-WJ).** When no jump episode is active, the chain transitions according to the empirical transition matrix $\mathbf{T}$ with probability $1-\epsilon$ (thin bidirectional arrows) or enters a Poisson jump episode with probability ϵ (dashed upward arrow to the Poisson clock). During an episode, each of the $K \sim \text{Poisson}(\lambda)$ forced steps independently draws the bottom tail set $\mathcal{S}_-$ (red, lowest-valued states) with probability p_{neg} or the top tail set $\mathcal{S}_+$ (blue, highest-valued states) otherwise. Ordinary transitions then resume. Small bell curves above each state show the state-conditional location-scale Student-t ($\nu = 5$) emission distribution. Tail sets are defined as the N_{tail} lowest- and highest-quantile states under the Laplace quantile partition.

We added jump episodes to extend visits to extreme-return states. When no jump episode was active, a new episode began with probability ϵ. Its duration was $K \sim \text{Poisson}(\lambda)$, where λ is the mean episode length (Glasserman, 2003). Because a Poisson draw can be zero, $K = 0$ caused no forced tail step. For $K > 0$, the model temporarily replaced ordinary transitions with draws from two tail sets. The set $\mathcal{S}_- = \{1, \ldots, N_{\text{tail}}\}$ contained the most negative states, and $\mathcal{S}_+ = \{N - N_{\text{tail}} + 1, \ldots, N\}$ contained the most positive states. At each forced step, the model selected the negative set with probability p_{neg} and the positive set otherwise. It made the tail-set choice independently at each step, so negative extreme returns occurred slightly more often than positive ones. Table 1 lists ϵ, λ, p_{neg}, and N_{tail}. We obtained the initial-state distribution $\bar{\pi}$ by solving $\bar{\pi}\mathbf{T} = \bar{\pi}$ with $\sum_k \bar{\pi}_k = 1$ (Hamilton, 2018; Grinstead and Snell, 1997). This distribution is exact for the ordinary transition matrix but only approximate for the jump-augmented process because it does not include the remaining jump duration. The full simulation procedure appears in Algorithm 1. Each step followed $\mathbf{T}$ or continued a jump episode and then drew a return from the selected state's emission distribution.

We chose the jump probability ϵ and mean duration λ to match the SPY absolute-return autocorrelation and kurtosis (Mandelbrot, 1963; Cont, 2001; Rydén et al., 1998). Jump-diffusion models use Poisson events to add instantaneous price jumps (Merton, 1976; Kou, 2002). In HMM-WJ, a jump episode extended a visit to the tail states. We searched a grid of (ϵ, λ) values. At each grid point, we averaged squared errors in the autocorrelation function (ACF) of absolute

Algorithm 1 Hidden Markov model with jumps (HMM-WJ) single-asset marginal generator.

Require: Fitted parameters $(\mathbf{T}, \mathbf{E}, \bar{\pi}, \epsilon, \lambda, p_{\text{neg}}, N_{\text{tail}})$; horizon T; tail sets $\mathcal{S}_- = \{1, \ldots, N_{\text{tail}}\}$ and $\mathcal{S}_+ = \{N - N_{\text{tail}} + 1, \ldots, N\}$.
Ensure: State sequence $\{S_t\}_{t=0}^{T}$ and growth-rate path $\{G_t\}_{t=1}^{T}$.

Draw $S_0 \sim \bar{\pi}$; $k \leftarrow 0$ {k: remaining forced-step counter}
for $t = 1, \ldots, T$ **do**
  if $k > 0$ **then**
    $\mathcal{J} \leftarrow \mathcal{S}_-$ w.p. p_{neg}, else $\mathcal{S}_+$; $S_t \sim \text{Uniform}(\mathcal{J})$; $k \leftarrow k - 1$ {forced tail step; sign drawn per step}
  else if $\text{Uniform}(0,1) > \epsilon$ **then**
    $S_t \sim \mathbf{T}_{S_{t-1},\cdot}$ {standard Markov transition}
  else
    $K \sim \text{Poisson}(\lambda)$ {trigger jump; K is the episode length}
    if $K = 0$ **then**
      $S_t \sim \mathbf{T}_{S_{t-1},\cdot}$ {$K = 0$: no forced steps, standard transition}
    else
      $\mathcal{J} \leftarrow \mathcal{S}_-$ w.p. p_{neg}, else $\mathcal{S}_+$; $S_t \sim \text{Uniform}(\mathcal{J})$; $k \leftarrow \min(K - 1, T - t)$ {first forced step; K steps total, sign per step}
    end if
  end if
  Draw $G_t \sim \mu_{S_t} + \sigma_{S_t}\, t_5$ {Eq. (2)}
end for

returns through lag L and weighted squared errors in excess kurtosis over the jump-active paths. We minimized the objective $J(\epsilon, \lambda)$.

$$J(\epsilon, \lambda) = \frac{1}{|\mathcal{R}_J|} \sum_{r \in \mathcal{R}_J} \left[\sum_{\tau=1}^{L} \left(\text{ACF}_{\text{obs}}(\tau) - \text{ACF}_r(\tau)\right)^2 + w_\kappa \left(\kappa_{\text{obs}} - \kappa_r\right)^2 \right], \quad (3)$$

where $\mathcal{R}_J$ is the set of jump-active simulated paths, $\text{ACF}_{\text{obs}}(\tau)$ and κ_{obs} are the empirical absolute-growth autocorrelation at lag τ and excess kurtosis, and $\text{ACF}_r(\tau)$ and κ_r are the corresponding values for path r. We used κ for kurtosis and K for jump duration. We simulated 200 independent paths of 2,766 trading days at each grid point. A path with no jump reduces to the hidden Markov model with no jumps (HMM-NJ) and contains no information about (ϵ, λ), so the objective included only paths with at least one jump (Glasserman, 2003). The value of w_κ set the relative weight of the kurtosis error. The search ranges, w_κ, L, and the remaining HMM-WJ and SIM hyperparameters are reported in Table 1. We fixed the selected hyperparameters before the generator comparisons and holdout evaluation in Section 4.

3.2. Variance-corrected multi-asset composition

The single-asset model generated each ticker separately and did not specify cross-asset dependence. For the multi-asset model, we added a simulated market path as a common factor. We used the single-index model (SIM) of Sharpe (1963).

$$g_i(t) \;=\; \alpha_i \;+\; \beta_i \, g_m(t) \;+\; \varepsilon_i(t), \qquad t = 1, \ldots, T. \tag{4}$$

Here α_i is the intercept for asset i, g_m is the market growth-rate path, β_i is the asset's market loading, and ε_i is its asset-specific residual path. We use ε_i for the residual path and ϵ for the jump probability in the single-asset model. We estimated the intercept and market loading (α_i, β_i) from real data by ordinary least squares (OLS). The choice of ε_i determined the composition method. A standard SIM draws independent Gaussian residuals with the variance estimated by OLS. These Gaussian residuals preserve the fitted market relationship but do not retain the heavy tails and volatility regimes produced by the single-asset generator. We called the Gaussian-residual method *Gaussian SIM*.

Naive full-path composition. In the naive construction, a full synthetic return path $\tilde{g}_i$ from the single-asset generator served as the residual.

$$g_i = \alpha_i + \beta_i \, g_m + \tilde{g}_i. \tag{5}$$

The fitted intercept and market term already set the conditional mean of asset i, while the full-return draw generally has $\mathbb{E}[\tilde{g}_i] \neq 0$. Equation (5) therefore has conditional mean $\alpha_i + \mathbb{E}[\tilde{g}_i] + \beta_i g_m$, not $\alpha_i + \beta_i g_m$.

We centered every full-return path before composition to remove the duplicate mean. Centering subtracted the realized path mean.

$$\bar{\tilde{g}}_i \equiv \frac{1}{T} \sum_{t=1}^{T} \tilde{g}_i(t), \qquad \tilde{g}_i^{\mathrm{c}}(t) \equiv \tilde{g}_i(t) - \bar{\tilde{g}}_i. \tag{6}$$

Therefore, $T^{-1} \sum_t \tilde{g}_i^{\mathrm{c}}(t) = 0$ exactly. Subtracting a constant leaves both the sample variance and the sample covariance with g_m unchanged. We used the centered construction $g_i = \alpha_i + \beta_i g_m + \tilde{g}_i^{\mathrm{c}}$ as the *naive* benchmark. Centering corrected the mean but not the variance.

The path variance $\sigma^2_{\mathrm{gen},i} = \mathrm{Var}(\tilde{g}_i) = \mathrm{Var}(\tilde{g}_i^{\mathrm{c}})$ is the *generator variance.* It already represents the asset's total variance. Adding the market term therefore adds variance a second time. We sampled the residual and market paths independently. Their covariance is zero in population and close to zero on a long realized path. We approximated the naive composed variance accordingly.

$$\mathrm{Var}(g_i) \;\approx\; \beta_i^2 \sigma_m^2 + \sigma^2_{\mathrm{gen},i}, \qquad \mathrm{Cov}(\tilde{g}_i^{\mathrm{c}}, g_m) \approx 0, \tag{7}$$

where $\sigma_m^2 = \mathrm{Var}(g_m)$ is the variance of the market path. The excess is $\beta_i^2 \sigma_m^2$, so the error is largest for assets with large market loadings.

Variance correction. We set $\sigma^2_{\mathrm{gen},i}$ as the target total variance for asset i. The market term contributed $\beta_i^2 \sigma_m^2$, so the residual term could contribute only the remaining variance. We defined s_i as the nonnegative asset-specific residual scale, set $\varepsilon_i = s_i \tilde{g}_i^{\mathrm{c}}$, and required $\mathrm{Var}(\beta_i g_m + s_i \tilde{g}_i^{\mathrm{c}}) = \sigma^2_{\mathrm{gen},i}$. Under the zero-covariance assumption, solving for s_i gives the correction.

$$\boxed{s_i^2 \;=\; 1 \;-\; \frac{\beta_i^2\,\sigma_m^2}{\sigma^2_{\mathrm{gen},i}}} \tag{8}$$

The *market loading ratio* ρ_i is the market variance contribution divided by the target variance.

$$\rho_i \;\equiv\; \frac{\beta_i^2\,\sigma_m^2}{\sigma^2_{\mathrm{gen},i}}, \tag{9}$$

Under the zero-covariance assumption, ρ_i is the fraction of the target variance contributed by the market, and $s_i^2 = 1 - \rho_i$. At $\rho_i = 1$, the unconstrained rule leaves no asset-specific variance; for $\rho_i > 1$, it has no real-valued solution. Values of ρ_i at or above one can arise when β_i is large, the generator variance is low, or the synthetic market is more volatile than the calibration market. We required the asset-specific draw to contribute at least a fraction f of the target variance. We set $\rho_i^{\max} = 1 - f$ and reduced the effective loading β_i^{eff} when $\rho_i > \rho_i^{\max}$. We applied the piecewise correction in Eq. (10).

$$\begin{aligned}
\beta_i^{\mathrm{eff}} \;&=\; \begin{cases} \beta_i & \text{if } \rho_i \le \rho_i^{\max}, \\[1ex] \operatorname{sign}(\beta_i)\,\sqrt{(1-f)\,\dfrac{\sigma^2_{\mathrm{gen},i}}{\sigma_m^2}} & \text{if } \rho_i > \rho_i^{\max}, \end{cases} \\
s_i^2 \;&=\; \begin{cases} 1 - \rho_i & \text{if } \rho_i \le \rho_i^{\max}, \\ f & \text{if } \rho_i > \rho_i^{\max}. \end{cases}
\end{aligned} \tag{10}$$

The correction retained the sign of the calibrated loading β_i. We used $f = 0.10$, so the asset-specific draw contributed at least 10% of the target variance for non-tracker assets (Table 1).

High-R^2 index trackers. The non-tracker branch targeted total generator variance. For an index-tracking fund, we instead targeted its fitted relationship with the market. The coefficient of determination R^2 measures the fraction of the asset's variance explained by the market. For example, SPY regressed on itself has $\beta = 1$, $R^2 = 1$, and no residual variance. QQQ and SPYG also had high R^2 against SPY. The non-tracker branch can change the fitted R^2 even when it preserves total variance. We therefore used a separate tracker branch when $R^2_{i,\mathrm{real}}$ was at least R^2_{preserve}. We set $R^2_{\mathrm{preserve}} = 0.80$, which separated the index trackers from individual stocks in our universe (Supplementary Fig. S3). For the tracker branch, we computed the residual variance required to recover the calibrated R^2 under the zero-covariance assumption.

$$\boxed{\sigma^2_{\varepsilon,\mathrm{target}} \;=\; \beta_i^2\,\sigma_m^2 \cdot \frac{1 - R^2_{i,\mathrm{real}}}{R^2_{i,\mathrm{real}}}} \tag{11}$$

Table 1: **Hidden Markov model with jumps (HMM-WJ) and single-index model (SIM) composition hyperparameters.** Values used for the main SPY analysis, explored ranges, and locations of sensitivity checks. For the 423-ticker SIM composition, we set $\epsilon = 0$ instead of the SPY value. The cross-asset validation used per-ticker (ϵ^*, λ^*) values for NVDA, JNJ, and JPM (Supplementary Table S7).

Symbol	Description	SPY value	Explored range
State partition and emission			
N	Laplace-quantile states	100	$\{30, \ldots, 350\}$, Online Appendix S6
ν	Student-t emission degrees of freedom	5	$\{3, \ldots, 30, \infty\}$
Jump-duration override			
ϵ	jump probability per step	10^{-4}	$[10^{-4}, 2.5 \times 10^{-2}]$, Eq. (3)
λ	mean jump-episode duration (steps)	90	$[10, 160]$, Eq. (3)
p_{neg}	probability a jump targets $\mathcal{S}_-$	0.52	fixed
N_{tail}	quantile states per tail set	5	fixed (at $N = 100$)
w_κ	kurtosis-penalty weight, Eq. (3)	0.20	fixed
L	autocorrelation lag window, Eq. (3)	25 days	fixed
SIM composition			
f	idiosyncratic variance floor	0.10	$\{0.05, \ldots, 0.30\}$
R^2_{preserve}	branch-selection threshold	0.80	$\{0.70, \ldots, 0.90\}$

We rescaled the generator draw to the target residual variance.

$$\varepsilon_i = \left(\sqrt{\frac{\sigma^2_{\varepsilon,\text{target}}}{\sigma^2_{\text{gen},i}}} \right) \tilde{g}_i^{\text{c}}. \tag{12}$$

When $R^2_{i,\text{real}} = 1$, the target residual variance is zero. SPY regressed on itself therefore has no residual term. A scale factor greater than one increases the variance of the generator draw and can weaken the original marginal fit.

Complete composition rule. The tracker branch targeted the calibrated R^2. The non-tracker branch targeted the generator variance and reduced β_i only when the market term would leave less than the required asset-specific variance. After selecting a branch, we added the scaled asset draw to the market term. The full procedure appears in Algorithm 2. If used, a residual-copula rank reorder was applied jointly after scaling and before composition. The reorder preserved the empirical variance of each residual path. Centering kept the path mean controlled by the intercept and market term. Under the zero-covariance assumption, OLS estimates from longer composed paths converge to the effective market loading and its intercept. The variance and loading properties are asymptotic. They do not specify exact finite-sample values or the complete return distribution. Online Appendix S8 gives the algebra. The construction adds only same-day market dependence; it does not reproduce lead–lag effects or market-driven volatility clustering.

Algorithm 2 Hybrid single-index model (SIM) composition with variance correction.

Require: Market growth-rate path $g_m \in \mathbb{R}^T$ with sample variance σ_m^2; per-asset generator growth-rate paths $\tilde{g}_i \in \mathbb{R}^T$ for $i = 1, \ldots, N$; calibrated SIM parameters $(\alpha_i, \beta_i, R^2_{i,\text{real}})$; idiosyncratic variance floor $f \in (0, 1)$; R^2-preservation threshold $R^2_{\text{preserve}} \in (0, 1)$.

Ensure: Composed paths $g_i \in \mathbb{R}^T$ and per-asset construction flags.

for each asset $i = 1, \ldots, N$ **do**
    Compute generator variance $\sigma^2_{\text{gen},i} \leftarrow \mathrm{Var}(\tilde{g}_i)$.
    Center the full-return draw: $\tilde{g}_i^{\text{c}} \leftarrow \tilde{g}_i - T^{-1} \sum_{t=1}^{T} \tilde{g}_i(t)$.
    if $R^2_{i,\text{real}} \geq R^2_{\text{preserve}}$ **then**
        $\sigma^2_{\varepsilon,\text{target}} \leftarrow \beta_i^2 \sigma_m^2 (1 - R^2_{i,\text{real}})/R^2_{i,\text{real}}$ {Eq. (11)}
        $\varepsilon_i \leftarrow \left(\sqrt{\sigma^2_{\varepsilon,\text{target}}/\sigma^2_{\text{gen},i}}\right) \tilde{g}_i^{\text{c}}$
        $\beta_i^{\text{eff}} \leftarrow \beta_i$; branch $\leftarrow$ R^2-preserving
    else
        $\rho_i \leftarrow \beta_i^2 \sigma_m^2 / \sigma^2_{\text{gen},i}$ {Eq. (9)}
        if $\rho_i \leq 1 - f$ **then**
            $s_i^2 \leftarrow 1 - \rho_i$; $\beta_i^{\text{eff}} \leftarrow \beta_i$; branch $\leftarrow$ variance-preserving
        else
            $s_i^2 \leftarrow f$; $\beta_i^{\text{eff}} \leftarrow \mathrm{sign}(\beta_i)\sqrt{(1-f)\,\sigma^2_{\text{gen},i}/\sigma_m^2}$; branch $\leftarrow$ clipped variance-preserving
        end if
        $\varepsilon_i \leftarrow s_i\, \tilde{g}_i^{\text{c}}$
    end if
    (Optional) apply cross-sectional rank reorder to ε_i.
    $g_i \leftarrow \alpha_i + \beta_i^{\text{eff}}\, g_m + \varepsilon_i$ {Eq. (4)}
end for

Evaluation protocol. We evaluated the method on 424 United States equities and exchange-traded funds from Polygon.io. Each ticker had 2,767 daily closing prices from January 2014 through December 2024, which yielded 2,766 growth rates. This complete-case selection conditioned the analysis on assets that survived and retained data coverage throughout the training window. We used SPY as the market and composed the other 423 assets. For each asset, we fitted the state and emission model used in the SPY analysis (Table 1; $r_f = 0$). We set the jump probability to zero so that the experiment measured multi-asset composition without per-ticker jump calibration. We regressed each real asset on SPY to estimate α_i, β_i, $R^2_{i,\text{real}}$, and residual variance. Across the sample, β_i ranged from 0.03 to 1.79, and $R^2_{i,\text{real}}$ ranged from 0.001 to 0.933 (Supplementary Fig. S3).

We compared six ways to construct the asset-specific term. The naive method used the full JumpHMM draw without correction. The Gaussian SIM method used independent Gaussian residuals with variance estimated by OLS. The hybrid method used Algorithm 2. The JumpHMM-on-residuals method fitted a new JumpHMM to the OLS residual series after converting the residuals to a synthetic price path by cumulative compounding. The block-bootstrap method sampled the OLS residual series with the Politis–Romano stationary bootstrap (Politis and Romano, 1994) and a mean block length of $\sqrt{T} \approx 50$. The GARCH(1,1)-t method fitted a conditional-variance model to each residual series (Bollerslev, 1986). We excluded thirty non-stationary GARCH fits from the GARCH results. For each ticker, the naive and hybrid methods received the same JumpHMM draw, which isolated the effect of the variance correction. The other four methods generated their own residual draws. We did not use the optional cross-sectional rank reorder because the experiment tested each ticker's marginal construction separately.

We evaluated each composed series on factor recovery, distributional fit, and variance preservation. For factor recovery, a new OLS regression produced the absolute errors $|\hat{\beta} - \beta|$ and $|\hat{R}^2 - R^2_{\mathrm{real}}|$; lower values indicate closer recovery. For distributional fit, we used Kolmogorov–Smirnov (KS) and Anderson–Darling (AD) pass rates, Wasserstein-1 distance, excess kurtosis, and the Hill upper-tail index. Higher KS and AD pass rates and lower Wasserstein distance indicate a closer match; kurtosis and the Hill index describe tail behavior. For variance preservation, we used the composed-to-generator variance ratio. A ratio of one indicates equal variances. Online Appendix S3 gives the metric definitions. We ran 100 replications per ticker with seed 1234, producing 42,300 series per method. Pass rates used the significance level $\alpha = 0.05$.

For the holdout evaluation, we froze all marginal fits, residual models, and SIM parameters estimated from 2014–2024. We evaluated the methods against the 249 growth rates observed in 2025. Of the 423 non-market training assets, 416 had complete holdout histories; GARCH results covered 386 because its thirty non-stationary training fits remained excluded. For each replication, we generated a 249-day SPY path from the training-period SPY marginal and used the same simulated market path in all asset compositions. Observed 2025 SPY and asset returns were used only for scoring. We repeated the KS, AD, Wasserstein-1, variance, kurtosis, and one-day left-tail Value-at-Risk (VaR) evaluations with 100 paths per ticker. For each VaR threshold, we also used Kupiec's unconditional-coverage test to compare the observed number of exceedances with the nominal rate (Kupiec, 1995). Because goodness-of-fit tests have less power on 249 observations than on 2,766, we also compared each synthetic path with a randomly selected contiguous 249-day block from the training history. We first averaged replication-level results within ticker and then summarized across tickers so each asset received equal weight. No 2025 data were used for fitting. The 249-day training blocks provided a comparison at the same sample length as the holdout period.

4. Results

4.1. Single-asset generator validation

We first tested whether the marginal generator in Fig. 2 reproduced the empirical SPY targets shown in Fig. 1. In both the in-sample (2014–2024,

Table 2: **In-sample and out-of-sample model comparison for SPY.** Results summarize 1,000 simulated paths. Values in parentheses are standard errors, and bold entries are best among the fitted generative models within each panel; the empirical bootstrap is excluded from this highlighting. Metric definitions and standard-error calculations are given in Online Appendix S3.

Model	KS (%)	AD (%)	κ	ACF-MAE	Coverage (%)	W_1	Hellinger
In sample: 2014–2024 *(*2,766 *days; observed* $\kappa = 7.715$*)*							
Bootstrap	100.0 (<0.1)	99.7 (0.2)	7.6 (0.08)	0.060 (<0.001)	100.0 (<0.1)	0.062 (0.001)	0.042 (<0.001)
Gaussian	0.0 (<0.1)	0.0 (<0.1)	−0.0 (<0.01)	0.060 (<0.001)	13.1 (0.1)	0.399 (0.001)	0.148 (<0.001)
Laplace	44.0 (1.6)	43.3 (1.6)	3.0 (0.02)	0.060 (<0.001)	37.4 (1.5)	0.138 (0.001)	**0.072** (<0.001)
GARCH(1,1)	5.5 (0.7)	1.9 (0.4)	8.2 (0.37)	**0.031** (0.001)	29.3 (1.4)	0.304 (0.006)	0.100 (0.001)
GRU	0.6 (0.2)	0.2 (0.1)	5.7 (0.16)	0.036 (<0.001)	17.2 (0.5)	0.421 (0.003)	0.134 (<0.001)
HSMM	82.0 (1.2)	42.5 (1.6)	4.8 (0.08)	0.059 (<0.001)	68.7 (1.1)	0.176 (0.001)	0.113 (<0.001)
HMM-NJ	**99.3** (0.3)	**98.2** (0.4)	8.1 (0.14)	0.059 (<0.001)	**100.0** (<0.1)	**0.081** (0.001)	**0.072** (<0.001)
HMM-WJ	98.3 (0.4)	94.8 (0.7)	**7.5** (0.09)	0.053 (<0.001)	**100.0** (<0.1)	0.097 (0.001)	0.074 (<0.001)
Out of sample: 2025 *(*249 *days; observed* $\kappa = 6.867$*)*							
Bootstrap	97.4 (0.5)	98.5 (0.4)	6.0 (0.18)	0.043 (<0.001)	**100.0** (<0.1)	0.232 (0.002)	0.205 (0.001)
Gaussian	62.2 (1.5)	46.3 (1.6)	−0.0 (<0.01)	0.043 (<0.001)	36.4 (1.9)	0.452 (0.002)	0.235 (0.001)
Laplace	88.0 (1.0)	92.9 (0.8)	2.7 (0.05)	0.043 (<0.001)	74.7 (1.0)	0.263 (0.002)	0.211 (0.001)
GARCH(1,1)	80.3 (1.3)	72.0 (1.4)	1.6 (0.07)	**0.026** (<0.001)	96.0 (1.3)	0.507 (0.018)	0.232 (0.001)
GRU	71.8 (1.4)	44.8 (1.6)	2.9 (0.10)	0.033 (<0.001)	77.8 (2.9)	0.488 (0.005)	0.240 (0.001)
HSMM	96.2 (0.6)	**96.7** (0.6)	4.1 (0.13)	0.042 (<0.001)	**100.0** (0.2)	0.287 (0.002)	0.239 (0.001)
HMM-NJ	**96.6** (0.6)	96.1 (0.6)	**6.2** (0.15)	0.041 (<0.001)	**100.0** (<0.1)	**0.259** (0.003)	**0.207** (0.001)
HMM-WJ	95.4 (0.7)	95.3 (0.7)	**6.2** (0.15)	0.040 (<0.001)	**100.0** (<0.1)	0.275 (0.005)	0.210 (0.001)

Model abbreviations are generalized autoregressive conditional heteroskedasticity (GARCH), gated recurrent unit (GRU), hidden semi-Markov model (HSMM), hidden Markov model with no jumps (HMM-NJ), and hidden Markov model with jumps (HMM-WJ). Kolmogorov–Smirnov (KS) and Anderson–Darling (AD) report pass rates at $\alpha = 0.05$; coverage is the fraction of empirical quantiles inside the synthetic 90% envelope. κ is simulated excess kurtosis. Mean absolute error in the autocorrelation function (ACF-MAE) is computed for $|G_t|$ through lag 252. Larger values are preferred for KS, AD, and coverage; smaller values are preferred for ACF-MAE, W_1, and Hellinger distance; κ is compared with the observed value.

$T = 2{,}766$) and out-of-sample (2025, $T = 249$) windows, the real SPY series had heavy tails, and the Jarque-Bera test rejected normality. The series also showed persistent volatility clustering. Individual raw-return correlations were small, although the Ljung–Box test detected joint serial dependence (Supplementary Table S1). We calibrated (ϵ, λ) for HMM-WJ against the in-sample SPY series using the grid search in Eq. (3) (Algorithm 1; search range in Table 1). The selected values were $(\epsilon, \lambda) = (10^{-4}, 90)$, although nearby grid points gave similar objective values. Under the counted transition matrix, ordinary state residence times were only 1–2 steps, whereas the selected jump duration had a mean of 90 steps (Supplementary Fig. S5). The primary analysis used $N = 100$ states. Across $N = 60$ to 200, the KS and AD pass rates varied by less than one percentage point for both HMM-NJ and HMM-WJ. At $N = 350$, some states were never visited and their emission parameters could not be estimated (Online Appendix S6). The results were stable across a broad range of state resolutions, so we kept $N = 100$ for the generator comparisons that followed.

With the model calibrated, we compared HMM-WJ with seven alternative generators on 1,000 paths of 2,766 trading days each (Table 2; Fig. 3). No baseline was best on both the return distribution and volatility clustering. Bootstrap resampling passed every distribution test but, like all independent sampling

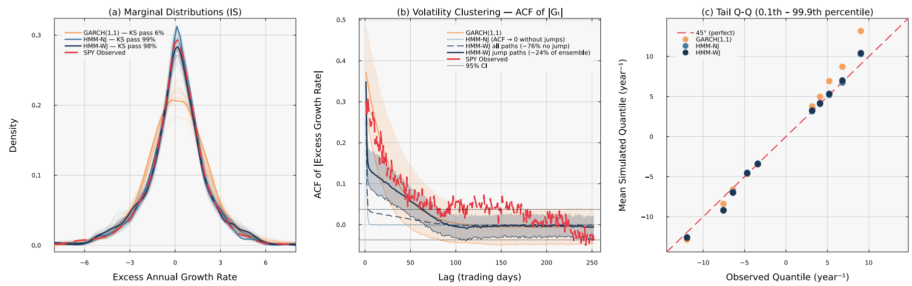

Figure 3: **Head-to-head in-sample model comparison for SPY** ($N = 100$, 1,000 simulated paths). *Panel (a):* marginal density of excess growth rates with in-sample Kolmogorov–Smirnov (KS) pass rates annotated per generator. *Panel (b):* autocorrelation function of $|G_t|$ at lags 1–252 (shaded bands: 10th–90th percentile across paths) per generator. *Panel (c):* tail quantile-quantile plot at the 0.1st–99.9th percentile region per generator.

methods, did not reproduce clustering. Gaussian sampling passed none of the distribution tests. The GARCH(1,1) model had the lowest mean absolute error in the autocorrelation function (ACF-MAE) of 0.031 but rarely passed the distribution tests. The gated recurrent unit (GRU) (Cho et al., 2014) showed the same general pattern. The HSMM-style baseline adapted from Bulla and Bulla (2006) reached an 82% KS pass rate but underestimated kurtosis by more than one third.

Among the fitted parametric and latent-state generators, the two HMM variants gave the strongest overall distributional fit. The HMM-NJ and HMM-WJ models had KS pass rates of 99.3% and 98.3%, respectively. The slightly lower HMM-WJ pass rates were consistent with jump episodes sometimes placing too much probability in the tails. Student-t emissions produced simulated kurtosis of 7.5, close to the observed 7.7, whereas Normal emissions underestimated it by about 29% (Supplementary Table S3; Online Appendix S5). Both HMM variants also reproduced the observed skewness of -0.75. That skewness arose from unequal time spent in negative and positive states rather than from an explicit skewness parameter.

The jump episodes improved temporal fit. The HMM-WJ model reduced ACF-MAE from 0.059 for HMM-NJ to 0.053. About 24% of HMM-WJ paths contained at least one jump and showed more persistent autocorrelation in absolute returns; paths without a jump were equivalent to HMM-NJ. On the held-out 249 days of 2025, HMM-NJ and HMM-WJ retained KS pass rates of 96.6% and 95.4%, with kurtosis close to the observed value (Table 2; Supplementary Fig. S4). Tests on NVDA, JNJ, and JPM showed strong in-sample fit but more variable holdout performance (Supplementary Table S7, Online Appendix S7). These comparisons support HMM-WJ as a compromise between distributional and temporal fit. We examined that compromise directly by varying the share of jump-active paths.

To measure how jump-active paths changed the results, we resampled the fitted ensemble at different jump-path fractions f (Fig. 4). Here f is the share

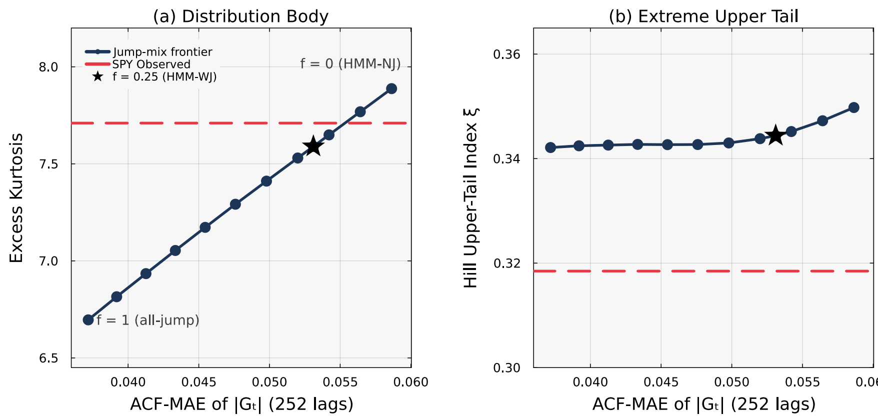

Figure 4: **Jump-mix enrichment frontier for SPY** (in-sample, $N = 100$, pool of 4,000 simulated paths). Resampling the fitted hidden Markov model with jumps (HMM-WJ) ensemble to a target jump-path fraction f (the share of accepted paths containing at least one jump) traces the temporal-marginal tradeoff. Both panels share the horizontal axis: mean absolute error in the autocorrelation function (ACF-MAE) for $|G_t|$ over lags 1–252); lower values indicate stronger volatility clustering. Panel (a): simulated excess kurtosis. Panel (b): Hill upper-tail index ξ, the raw estimator equal to the reciprocal of the tail exponent. Red dashed lines mark the observed SPY values; the star marks the $f \approx 0.25$ value used throughout. Endpoints $f = 0$ and $f = 1$ correspond to the no-jump generator and an all-jump ensemble.

of accepted paths that contain at least one jump. We varied f from 0, where all accepted paths reduce to HMM-NJ, to 1, where every accepted path contains a jump. This procedure held the fitted model fixed and changed only the mixture of accepted paths. The absolute-return autocorrelation error fell monotonically from 0.059 at $f = 0$ to 0.037 at $f = 1$, moving from the no-jump value toward the GARCH value as the jump share rose. Over the same sweep, simulated excess kurtosis declined from 7.9 to 6.7, matching the observed value near $f \approx 0.14$ and falling below it thereafter. The two-sample KS and AD pass rates also declined. The Hill upper-tail index stayed near 0.35, against an observed 0.32, across the whole sweep because the Student-t ($\nu = 5$) emission sets the tail exponent independently of jump content. Increasing the jump share therefore traded fit to the distributional body for lower autocorrelation error without changing the tail index.

The value $f \approx 0.25$ used throughout kept simulated kurtosis within 2% of the observed value and lowered the autocorrelation error about a quarter of the way from the no-jump value toward the all-jump endpoint. The selected mixture therefore provided a compromise between marginal and temporal fit before the fitted marginals were carried into the multi-asset composition experiment.

4.2. Variance-corrected multi-asset composition

The single-asset results defined the tradeoff in the marginal generator. We then tested whether the variance-corrected SIM preserved each asset's variance and factor loading across the 423-ticker universe, comparing the six composition methods on the aggregate scorecard (Table 3). Before comparing the methods, we measured the cross-term omitted from the variance correction. For the full-return generator, the universe-mean normalized cross-covariance with the market path was 1.53×10^{-4}, and the per-ticker values ranged from -0.0061 to 0.0061 (Supplementary Fig. S1). These values supported the negligible-cross-term approximation used in Eq. (8).

Table 3: **Aggregate scorecard across 423 non-market tickers and 100 replications per ticker.** Errors, distances, and tail metrics are medians per method. Kolmogorov–Smirnov (KS) and Anderson–Darling (AD) pass fractions are pooled across replications. $|\hat{\alpha} - \alpha|$, $|\hat{\beta} - \beta|$, and $|\hat{R}^2 - R^2_{\text{real}}|$: absolute deviation between ordinary least-squares (OLS) recovery and calibration (intercept error has units of yr^{-1}). KS pass and AD pass: fraction of replications whose two-sample test against the real marginal exceeds $p = 0.05$. W_1: Wasserstein-1 distance to the real marginal. κ: sample excess kurtosis. Hill: tail-index estimate on the upper 5% of $|g|$. SIM denotes the single-index model, and GARCH denotes generalized autoregressive conditional heteroskedasticity.

Method	$\lvert\hat{\alpha} - \alpha\rvert$	$\lvert\hat{\beta} - \beta\rvert$	$\lvert\hat{R}^2 - R^2_{\text{real}}\rvert$	KS pass (%)	AD pass (%)	W_1	κ	Hill
Naive	0.002	0.022	0.087	7.1	3.5	0.700	7.767	0.369
Gaussian SIM	0.048	0.017	0.008	0.6	0.5	0.682	1.427	0.217
Hybrid	0.002	0.018	0.021	73.6	64.6	0.249	7.165	0.365
JumpHMM-on-residuals	0.049	0.018	0.015	75.8	70.7	0.232	7.262	0.365
Block bootstrap	0.042	0.017	0.020	73.3	66.8	0.232	6.675	0.330
GARCH(1,1)-t	0.047	0.017	0.036	67.4	61.4	0.267	6.069	0.317

All six methods recovered the factor loading with similar accuracy: the median absolute β error ranged from 0.017 to 0.022 (Table 3). Path centering also kept the median absolute intercept error at $0.002\,\text{yr}^{-1}$ for both methods that used full-return draws. The residual-fit methods retained random variation in the residual mean over each finite path. Their larger intercept errors therefore do not, by themselves, imply a worse fit to the return distribution.

The larger differences appeared in variance recovery and tail behavior. Gaussian SIM had the lowest median R^2 error because it targets R^2 directly. However, it almost never passed the distributional tests, and its median excess kurtosis was an order of magnitude below the observed value (Supplementary Fig. S2). Centered naive composition retained the shape of the full-return draw, but adding the market factor inflated its variance by $\beta_i^2 \sigma_m^2$. The hybrid method removed that extra variance. It passed 73.6% of KS comparisons and retained heavy tails, with median excess kurtosis of 7.2 and a Hill index of 0.36. Its median R^2 error fell between those of the naive and Gaussian methods because this branch preserves the generator variance rather than targeting the observed R^2 directly.

The three residual-fit methods avoided double counting by fitting their generators to the empirical OLS residuals, whose variance already excludes the market component. Their KS pass rates ranged from 67.4% for GARCH(1,1)-t to 75.8% for JumpHMM-on-residuals. The hybrid rate was two percentage points below JumpHMM-on-residuals and slightly above block bootstrap. JumpHMM-on-residuals and block bootstrap had slightly lower Wasserstein distances. JumpHMM-on-residuals also matched the hybrid's kurtosis and Hill index almost exactly, whereas the block-bootstrap and GARCH residual fits produced lighter tails. The correction substantially improved on naive composition and approached the best residual-fit method without requiring a second fit for each asset. We next tested whether this ordering persisted after all fitted parameters were frozen.

We froze the fitted models and SIM parameters before evaluating them against 416 complete asset histories from 2025 (Table 4). The hybrid method reached an 87.0% mean cross-ticker KS pass rate, against 83.5% for centered naive composition, and its median composed-to-observed variance ratio was 0.97, whereas naive composition remained inflated at 1.29. JumpHMM-on-residuals reached an 86.2% KS pass rate and a slightly lower median W_1 than the hybrid. Matched-length training blocks produced lower pass rates for every method. The higher 2025 pass rates did not arise solely from comparing samples of 249 rather than 2,766 observations.

The average pass rates concealed substantial differences among tickers (Fig. 5). Most ticker-level rates were near the upper end of the range, but every method retained a lower tail of weak fits. For the hybrid method, many tickers improved relative to their matched-length training blocks, while a subset deteriorated out of sample. The held-out marginal improvement was broad but not uniform across assets, motivating a separate check of whether the resulting synthetic tails produced reliable risk thresholds.

Table 4: **Frozen-parameter multi-asset evaluation on the 2025 holdout.** Results cover 416 non-market tickers and 100 replications per ticker; generalized autoregressive conditional heteroskedasticity (GARCH) covers 386 tickers. Pass rates are first averaged within ticker and then across tickers. The *matched in-sample (IS)* columns compare each synthetic path with a randomly selected contiguous 249-day block from the 2014–2024 training history, matching the holdout sample length. W_1 and the variance ratio are medians across ticker-level medians; the variance ratio is synthetic variance divided by observed 2025 variance.

	KS pass (%)		AD pass (%)			
Method	matched IS	2025	matched IS	2025	W_1 (2025)	variance ratio
Naive	69.4	83.5	54.8	71.6	0.877	1.29
Gaussian SIM	57.4	73.9	44.9	64.6	0.866	0.97
Hybrid	78.8	87.0	66.2	79.7	0.716	0.97
JumpHMM-on-residuals	77.9	86.2	65.6	80.1	0.708	0.98
Block bootstrap	76.3	84.0	64.0	76.5	0.715	0.90
GARCH(1,1)-t	74.5	82.6	62.6	75.0	0.733	0.91

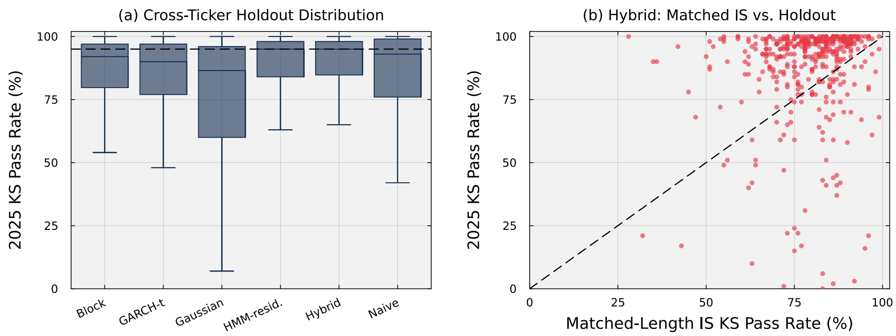

Figure 5: **Cross-ticker distribution of frozen-parameter out-of-sample marginal fit.** Panel (a) shows the distribution of per-ticker 2025 Kolmogorov–Smirnov (KS) pass rates across 100 replications; the dashed line marks 95%. Panel (b) compares each ticker's hybrid KS pass rate against a matched-length training block with its pass rate against the 2025 holdout. The diagonal denotes equal performance.

4.3. VaR coverage

We used the frozen synthetic paths to estimate per-ticker one-day left-tail Value-at-Risk (VaR) thresholds and counted exceedances on the real 2025 histories (Table 5). At $\alpha = 0.95$, the hybrid and JumpHMM-on-residuals methods produced mean exceedance rates of 5.97% and 6.07%, respectively, against the 5% nominal rate. At $\alpha = 0.99$, both rates were 1.67%, against the 1% nominal rate. Naive composition was closest at the 99% level (1.21%), while Gaussian SIM had the largest exceedance rate (2.03%). The short holdout contains only about 2–3 expected 99% exceedances per ticker, so we interpret the corresponding Kupiec pass rates and cross-ticker dispersion as low-power checks rather than precise rankings. Taken together, the VaR results showed that stronger marginal fit did not automatically imply nominal tail coverage, motivating a closer look at where the variance correction was most stable.

Table 5: **Out-of-sample per-ticker one-day left-tail Value-at-Risk coverage across the six composition methods.** VaR thresholds at coverage level α were estimated from frozen-parameter synthetic paths and exceedances were counted against the real 2025 growth-rate history. *rate*: mean of the per-ticker mean exceedance rates (nominal: $1 - \alpha$). Cross-ticker standard deviation (SD) is reported in percentage points. *Kupiec pass*: mean within-ticker fraction of replications whose unconditional-coverage test p-value exceeds 0.05.

	$\alpha = 0.95$			$\alpha = 0.99$		
Method	rate (%)	SD (pp)	Kupiec pass (%)	rate (%)	SD (pp)	Kupiec pass (%)
Naive	4.37	2.15	65.8	1.21	0.76	78.9
Gaussian SIM	4.95	2.26	70.0	2.03	1.17	72.1
Hybrid	5.97	2.43	66.3	1.67	0.93	73.4
JumpHMM-on-residuals	6.07	2.59	67.0	1.67	0.95	74.5
Block bootstrap	6.13	2.56	63.7	1.84	1.01	73.4
GARCH(1,1)-t	5.79	2.47	62.4	1.80	1.02	72.0

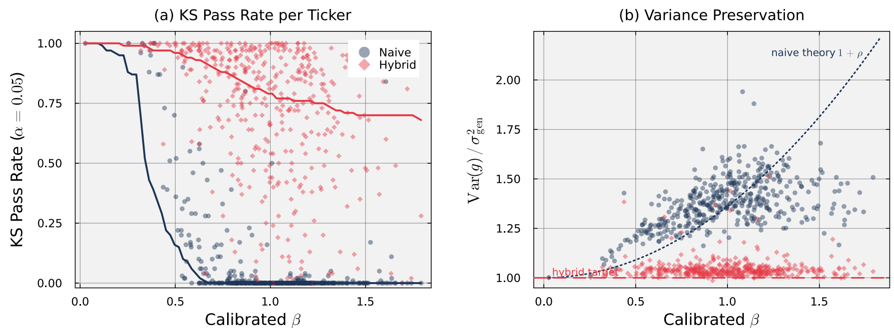

Figure 6: **Effect of the hybrid variance correction on each ticker's Kolmogorov–Smirnov pass rate and variance ratio.** Each point is a per-ticker median over 100 replications; thick lines are binned medians. *Panel (a):* Kolmogorov–Smirnov pass rate against the real marginal at $\alpha = 0.05$ for naive composition (blue) and hybrid (orange). *Panel (b):* variance ratio $\mathrm{Var}(g)/\sigma^2_{\mathrm{gen}}$ for the same two methods. The dotted curve is the theoretical naive inflation $1+\rho$ evaluated at the universe-median σ^2_{gen}; the dashed line marks a variance ratio of one.

4.4. Sensitivity analyses

To isolate how the variance correction changed marginal fit, we plotted each ticker's KS pass rate and variance ratio $\mathrm{Var}(g)/\sigma^2_{\mathrm{gen}}$ against its calibrated β (Fig. 6). Under naive composition, the variance ratio followed $1+\rho_i$ and exceeded generator variance by 50–75% at high β, and the naive KS pass rate fell toward zero above $\beta \approx 0.7$; the hybrid method held the ratio at one and retained substantially higher pass rates across the β range.

The branch-selection rule assigned 421 tickers to the variance-preserving branch, two trackers to the R^2-preserving branch, and none to the clipped branch (Supplementary Figs. S3 and S6; Supplementary Table S4). The two R^2-preserving assignments were QQQ and SPYG, and both had a 100% KS pass rate; the highest-R^2 individual stock, BLK, fell about 0.16 below the $R^2_{\mathrm{preserve}} = 0.80$ threshold. Clipping did not occur in the universe: $\rho_i = \beta_i^2\sigma_m^2/\sigma^2_{\mathrm{gen},i}$ (Eq. (9)) stayed below $1-f = 0.90$ for every ticker. Because the R^2-preserving branch contained only two tickers, we tested it on a synthetic-tracker grid spanning $(\beta_{\mathrm{true}}, R^2_{\mathrm{true}}) \in \{0.8, 1.0, 1.2\} \times \{0.80, 0.85, 0.90, 0.95, 0.99\}$. Each cell had a 100% KS pass rate across replications. The correction removed the observed beta-related variance inflation. We then tested the clipping branch under conditions that would force it to activate.

Bucketing the universe by β quartile, the hybrid KS pass rate fell from 87.6% in the lowest quartile to 65.9% in the highest (Supplementary Table S5). As β increased, the market term accounted for more of the composed variance, making the marginal more dependent on the market distribution. We tested the clipping rule directly by rescaling the market growth-rate path by $\gamma \in \{1, 2, 3\}$, holding the calibrated marginals fixed (Supplementary Table S6). At $\gamma = 1$ no ticker crossed the threshold. At $\gamma = 2$ and 3, clipping applied to 76.4% and 95.2%

of tickers. The clipping rule was unnecessary under observed market variance but activated as intended when market volatility was artificially increased. At the observed market variance, the floor f did not affect any branch assignment; varying R^2_{preserve} changed only the two tracker assignments identified above.

4.5. Daily-rebalanced portfolio check

To test a use that depends on joint cross-asset behavior, we applied a 20-asset target-weight allocator to the hybrid-composed 2025 paths. Switching the same target weights from static allocation to daily rebalancing increased the median annualized growth rate from 1.97% to 29.69%. Adding the allocation signal increased it by only another 0.13 percentage points, to 29.83%. The same allocator on the real 2025 history returned 11.50% (Supplementary Table S2). This comparison was consistent with omitted residual covariance creating a diversification-return bias, but it did not isolate how much of the gap that omission caused. A constant log-wealth shift set the synthetic ensemble median to the prespecified 8% annual return prior, but it did not restore the missing covariance.

5. Discussion

The single-asset analysis supplied the return generator used in the multi-asset construction. Relative to HMM-NJ, HMM-WJ reduced the autocorrelation error for absolute returns, although its in-sample KS and AD pass rates were slightly lower. The GARCH model achieved the lowest autocorrelation error of any generator but rarely passed the distributional tests, and the GRU showed a similar tradeoff. The adapted HSMM-style baseline passed the KS test more often but produced too little kurtosis. Bootstrap resampling passed the distributional tests by drawing directly from the observed returns, but it did not reproduce volatility clustering. We therefore interpret HMM-WJ as a compromise between distributional and temporal fit, not as a uniformly better single-asset model.

The distributional pass rates also need cautious interpretation. The two-sample KS and AD tests assume independent observations, whereas the HMM, HSMM, GARCH, and block-bootstrap paths contain dependence. For that reason, we considered the tests alongside Wasserstein and Hellinger distances and the Hill tail index. Model resolution creates a second qualification. A sensitivity sweep over $N \in \{30, 60, 90, 100, 150, 200\}$ showed little change across the middle of this range for the 2,766-day SPY sample, but shorter histories will support fewer well-populated states. We also tuned the jump parameters (ϵ, λ) on SPY and then applied them across the asset universe. Separate fits for NVDA, JNJ, and JPM achieved in-sample KS pass rates of at least 98.9%, but three assets cannot establish that one jump calibration transfers to every return series. Finally, the Student-t emissions model heavy tails but not skewness directly. The fitted skewness instead came from unequal time spent in negative and positive states, so assets with different asymmetry may need a different state partition.

The main multi-asset result was the variance correction. Naive composition added market variance to a draw that already represented the asset's total variance. The correction removed that double count and passed 73.6% of the full-length in-sample KS comparisons. After all fits were frozen, it raised the mean cross-ticker KS pass rate from 83.5% under centered naive composition to 87.0% on the 2025 holdout and kept the median variance ratio at 0.97. The improvement was broad but not uniform: every method retained a lower tail of tickers with weak fits, and some hybrid fits deteriorated relative to matched-length training blocks. A high average pass rate is therefore not a guarantee for every asset.

Fitting the return generator directly to OLS residuals is a competitive alternative. Residual-fit methods avoid double counting because the market component has already been removed. The best such method, JumpHMM-on-residuals, had a slightly higher KS pass rate in sample, while the hybrid method had the higher rate in the holdout. The practical difference is reuse: the hybrid construction can use a generator already fitted to full returns, whereas the residual method requires another fit for each asset. Neither approach achieved nominal tail coverage in the short holdout. The hybrid method's one-day left-tail 99% VaR exceedance rate was 1.67%, the same rounded rate as JumpHMM-on-residuals and above the 1% nominal rate.

Several design choices narrow this comparison. In sample, we used the observed SPY history as the market path. In the holdout experiment, we generated SPY from its frozen training-period model and reserved the observed 2025 path for scoring. The naive and hybrid methods shared the same JumpHMM draw for each ticker, while the residual methods used draws from their own fitted generators. We omitted the optional cross-sectional rank reorder and evaluated VaR for each ticker rather than for a portfolio. The variance correction also assumes $\mathrm{Cov}(\tilde{g}_i^c, g_m) \approx 0$, which holds here because the asset and market paths were generated independently. A generator sampled from joint market history would require a covariance adjustment. In addition, the present construction adds only contemporaneous market dependence; it does not reproduce lead-lag effects or market-driven volatility clustering.

The sensitivity results mark the boundary of the observed evidence. Hybrid KS pass rates declined from 87.6% in the lowest β quartile to 65.9% in the highest because the market path determined more of the composed distribution as β increased. The clipping rule did not activate in the observed universe, where every ticker had $\rho_i < 0.90$, but it activated for most tickers after market volatility was doubled or tripled. Only two tracker assets used the R^2-preserving branch. The 249-day holdout also had much less power than the 2,766-day training comparison. Matched-length training blocks partly address that difference, but one year of test data cannot provide a precise ranking from either pass rates or left-tail 99% VaR counts. The evidence therefore supports per-asset variance correction and distributional evaluation, not portfolio-level dependence fidelity.

The portfolio experiment shows why that distinction matters. Independent residual draws removed the sector and style covariance present in real assets (Booth and Fama, 1992). Daily rebalancing then converted the excess cross-sectional dispersion into a spurious return premium. This problem affected every method that generated residuals independently, including the block-bootstrap and GARCH variants. In a 20-asset target-weight example, the synthetic median annualized growth rate exceeded the realized 2025 return by about 18 percentage points. Moving from static weights to daily rebalancing changed the median by 27.73 percentage points, whereas adding the allocation signal changed it by only 0.13 percentage points (Booth and Fama, 1992; Bouchey et al., 2015; Markowitz, 1976). The apparent allocation gain was therefore almost entirely a diversification-return bias.

A joint residual model could address this limitation. A synchronized multivariate block bootstrap, residual factor model, or copula could restore cross-asset residual covariance. Multi-factor or lagged-factor versions could also capture dependence omitted by the current single-factor construction, with the variance correction retained as a compositional constraint. Per-asset jump calibration and stress tests involving highly volatile markets or leveraged assets would test the other transfer limits identified here. Separate univariate autoregressive or block-bootstrap models would not be enough: they can improve temporal behavior within each asset without creating dependence between assets.

The current location adjustment does not resolve this central limitation. We calibrated a constant shift in log-wealth to an external 8% annual long-run return prior (Damodaran, 2025; Ibbotson and Duff and Phelps, 2024), not to the realized test-year result. The shift preserved path ranks and adjusted the return level, although it changed drawdown timing and depth as wealth accumulated. It did not restore the missing residual covariance. The variance correction is therefore necessary for combining full-return generators with a market factor, but it is not a complete dependence model. Portfolio-level use requires joint residual dynamics.

6. Conclusion

In this study, we developed a variance-corrected method for composing per-asset generators through a single-index model. Centering first prevented a full-return draw from shifting the calibrated intercept; the subsequent scale correction removed the variance inflation caused by adding the market factor. Across 423 non-market assets, it recovered the calibrated factor loading to a low median absolute error and produced heavy-tailed marginals. On 416 complete held-out asset histories from 2025, it improved the mean cross-ticker KS pass rate from 83.5% under centered naive composition to 87.0% and held the median variance ratio at 0.97. Its one-day left-tail 99% Value-at-Risk exceedance rate was 1.67%, the same rounded rate as JumpHMM-on-residuals but above the 1% nominal rate. Those residual models, fitted directly to the OLS residual series, required a separate fit for each asset and produced a similar marginal pass rate.

The per-asset component underpinning this construction was a hidden Markov marginal model with Laplace-quantile states, Student-t emissions, and direct transition counting. On SPY, its jump-duration variant reduced absolute-return autocorrelation error relative to the no-jump variant while keeping high KS and AD pass rates; GARCH reached a lower autocorrelation error but failed the distributional tests. The single-asset result therefore showed a measured tradeoff between distributional fit and volatility clustering, not uniform superiority. Taken together, these results separate two tasks that are easily conflated when synthesizing multi-asset market data: the marginal generator governs each asset's distributional and temporal behavior, whereas the variance correction prevents the common market factor from double counting the generator's total variance.

The current multi-asset implementation does not model cross-asset residual covariance. In a 20-asset allocator, independent residuals inflated the synthetic median daily-rebalanced return well above the realized 2025 outcome, almost entirely through the static-to-daily-rebalance diversification return rather than the allocation signal. A location-shift correction set to a documented 8% annual long-run prior adjusted the return level without restoring the missing covariance. A full joint residual model is needed before using these paths for portfolio-level return forecasts. The variance correction is a necessary compositional constraint rather than a complete dependence model. Portfolio-level synthetic market data would require combining it with joint residual dynamics.

Code and data availability

The experiment scripts, evaluation code (six-method comparison, left-tail Value-at-Risk coverage check, and bias correction), cached inputs and result summaries, and fitted per-asset marginals are publicly available. The underlying model-fitting library is released separately and pinned as a dependency of the experiment code. Identifying repository links have been withheld from this manuscript for double-anonymized review and are provided to the editorial office on the separate title page.

Declaration of generative artificial intelligence use in the writing process

During the preparation of this work, the authors used OpenAI Codex to assist with language editing, consistency checks, and LaTeX formatting. The authors reviewed and edited all output and take full responsibility for the content of the publication.

References

Aas, K., Czado, C., Frigessi, A., Bakken, H., 2009. Pair-copula constructions of multiple dependence. Insurance: Mathematics and Economics 44, 182–198.

Assefa, S.A., Dervovic, D., Mahfouz, M., Tillman, R.E., Reddy, P., Veloso, M., 2020. Generating synthetic data in finance: Opportunities, challenges and pitfalls, in: Proceedings of the First ACM International Conference on AI in Finance, pp. 1–8. doi:10.1145/3383455.3422554.

Bauwens, L., Laurent, S., Rombouts, J.V.K., 2006. Multivariate GARCH models: A survey. Journal of Applied Econometrics 21, 79–109. doi:10.1002/jae.842.

Bilmes, J.A., 1998. A Gentle Tutorial of the EM Algorithm and Its Application to Parameter Estimation for Gaussian Mixture and Hidden Markov Models. Technical Report TR-97-021. International Computer Science Institute.

Bollerslev, T., 1986. Generalized autoregressive conditional heteroskedasticity. Journal of Econometrics 31, 307–327. doi:10.1016/0304-4076(86)90063-1.

Booth, D.G., Fama, E.F., 1992. Diversification returns and asset contributions. Financial Analysts Journal 48, 26–32. doi:10.2469/faj.v48.n3.26.

Bouchey, P., Nemtchinov, V., Wong, T.K.L., 2015. Volatility harvesting in theory and practice. Journal of Wealth Management 18, 89–100. doi:10.3905/jwm.2015.18.3.089.

Bulla, J., Bulla, I., 2006. Stylized facts of financial time series and hidden semi-Markov models. Computational Statistics & Data Analysis 51, 2192–2209. doi:10.1016/j.csda.2006.07.021.

Cherubini, U., Luciano, E., Vecchiato, W., 2004. Copula Methods in Finance. John Wiley & Sons.

Cho, K., van Merriënboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., Bengio, Y., 2014. Learning phrase representations using RNN encoder-decoder for statistical machine translation, in: Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP), Association for Computational Linguistics. pp. 1724–1734. doi:10.3115/v1/D14-1179.

Cont, R., 2001. Empirical properties of asset returns: Stylized facts and statistical issues. Quantitative Finance 1, 223–236. doi:10.1080/713665670.

Damodaran, A., 2025. Equity risk premiums: Determinants, estimation and implications. NYU Stern School of Business working paper, updated annually. https://pages.stern.nyu.edu/~adamodar/.

Dias, J.G., Vermunt, J.K., Ramos, S., 2010. Mixture hidden Markov models in finance research, in: Advances in Data Analysis, Data Handling and Business Intelligence, Springer. pp. 451–459. doi:10.1007/978-3-642-01044-6_41.

Embrechts, P., McNeil, A.J., Straumann, D., 2002. Correlation and dependence in risk management: Properties and pitfalls, in: Risk Management: Value at Risk and Beyond. Cambridge University Press, pp. 176–223. doi:10.1017/CBO9780511615337.008.

Engle, R., 2002. Dynamic conditional correlation: A simple class of multivariate generalized autoregressive conditional heteroskedasticity models. Journal of Business & Economic Statistics 20, 339–350. doi:10.1198/073500102288618487.

Engle, R.F., 1982. Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. Econometrica 50, 987–1007.

Fama, E.F., French, K.R., 1993. Common risk factors in the returns on stocks and bonds. Journal of Financial Economics 33, 3–56. doi:10.1016/0304-405X(93)90023-5.

Glasserman, P., 2003. Monte Carlo Methods in Financial Engineering. Springer. doi:10.1007/978-0-387-21617-1.

Grinstead, C.M., Snell, J.L., 1997. Introduction to Probability. American Mathematical Society.

Hamilton, J.D., 1989. A new approach to the economic analysis of nonstationary time series and the business cycle. Econometrica 57, 357–384. doi:10.2307/1912559.

Hamilton, J.D., 2018. Regime switching models. The New Palgrave Dictionary of Economics , 11421–11426doi:10.1057/978-1-349-95189-5_2459.

Hamilton, J.D., Susmel, R., 1994. Autoregressive conditional heteroskedasticity and changes in regime. Journal of Econometrics 64, 307–333. doi:10.1016/0304-4076(94)90067-1.

Heston, S.L., 1993. A closed-form solution for options with stochastic volatility with applications to bond and currency options. The Review of Financial Studies 6, 327–343. doi:10.1093/rfs/6.2.327.

Ibbotson, R.G., Duff and Phelps, 2024. SBBI Yearbook: Stocks, Bonds, Bills, and Inflation. Annual ed., Duff and Phelps / Kroll.

Jordon, J., Szpruch, L., Houssiau, F., Bottarelli, M., Cherubin, G., Maple, C., Cohen, S.N., Weller, A., 2022. Synthetic data: What, why and how? arXiv preprint arXiv:2205.03257 .

Kim, S., Shephard, N., Chib, S., 1998. Stochastic volatility: Likelihood inference and comparison with ARCH models. The Review of Economic Studies 65, 361–393.

Klaassen, F., 2002. Improving GARCH volatility forecasts with regime-switching GARCH. Empirical Economics 27, 363–394. doi:10.1007/s001810100100.

Kotz, S., Kozubowski, T.J., Podgórski, K., 2001. The Laplace Distribution and Generalizations: A Revisit with Applications to Communications, Economics, Engineering, and Finance. Birkhauser.

Kou, S.G., 2002. A jump-diffusion model for option pricing. Management Science 48, 1086–1101. doi:10.1287/mnsc.48.8.1086.166.

Kupiec, P.H., 1995. Techniques for verifying the accuracy of risk measurement models. The Journal of Derivatives 3, 73–84. doi:10.3905/jod.1995.407942.

Kwon, S., Lee, Y., 2024. Can GANs learn the stylized facts of financial time series?, in: Proceedings of the 5th ACM International Conference on AI in Finance, pp. 126–133. doi:10.1145/3677052.3698661.

Mandelbrot, B., 1963. The variation of certain speculative prices. The Journal of Business 36, 394–419.

Markowitz, H.M., 1976. Investment for the long run: New evidence for an old rule. Journal of Finance 31, 1273–1286. doi:10.1111/j.1540-6261.1976.tb03213.x.

Merton, R.C., 1976. Option pricing when underlying stock returns are discontinuous. Journal of Financial Economics 3, 125–144. doi:10.1016/0304-405X(76)90022-2.

Ni, H., Szpruch, L., Sabate-Vidales, M., Xiao, B., Wiese, M., Liao, S., 2021. Sig-wasserstein GANs for time series generation, in: Proceedings of the Second ACM International Conference on AI in Finance, ACM. pp. 1–8. doi:10.1145/3490354.3494393.

Patton, A.J., 2006. Modelling asymmetric exchange rate dependence. International Economic Review 47, 527–556. doi:10.1111/j.1468-2354.2006.00387.x.

Politis, D.N., Romano, J.P., 1994. The stationary bootstrap. Journal of the American Statistical Association 89, 1303–1313. doi:10.1080/01621459.1994.10476870.

Rabiner, L., Juang, B., 1986. An introduction to hidden Markov models. IEEE ASSP Magazine 3, 4–16. doi:10.1109/MASSP.1986.1165342.

Rizzato, M., Wallart, J., Geissler, C., Morizet, N., Boumlaik, N., 2023. Generative adversarial networks applied to synthetic financial scenarios generation. Physica A: Statistical Mechanics and its Applications 623, 128899. doi:10.1016/j.physa.2023.128899.

Ross, S.A., 1976. The arbitrage theory of capital asset pricing. Journal of Economic Theory 13, 341–360. doi:10.1016/0022-0531(76)90046-6.

Rydén, T., Teräsvirta, T., Åsbrink, S., 1998. Stylized facts of daily return series and the hidden Markov model. Journal of Applied Econometrics 13, 217–244. doi:10.1002/(SICI)1099-1255(199805/06)13:3<217::AID-JAE476>3.0.CO;2-V.

Sharpe, W.F., 1963. A simplified model for portfolio analysis. Management Science 9, 277–293. doi:10.1287/mnsc.9.2.277.

Stein, E.M., Stein, J.C., 1991. Stock price distributions with stochastic volatility: An analytic approach. The Review of Financial Studies 4, 727–752.

Stenger, M., Bauer, A., Prantl, T., Leppich, R., Hudson, N., Chard, K., Foster, I., Kounev, S., 2024. Thinking in categories: A survey on assessing the quality for time series synthesis. Journal of Data and Information Quality 16, 1–32. doi:10.1145/3666006.

Takahashi, S., Chen, Y., Tanaka-Ishii, K., 2019. Modeling financial time-series with generative adversarial networks. Physica A: Statistical Mechanics and Its Applications 527, 121261. doi:10.1016/j.physa.2019.121261.

Toth, D., Jones, B., 2019. Against the norm: Modeling daily stock returns with the Laplace distribution. arXiv preprint arXiv:1906.10325 .

Wiese, M., Knobloch, R., Korn, R., Kretschmer, P., 2020. Quant GANs: Deep generation of financial time series. Quantitative Finance 20, 1419–1440. doi:10.1080/14697688.2020.1730426.

Yoon, J., Jarrett, D., van der Schaar, M., 2019. Time-series generative adversarial networks, in: Advances in Neural Information Processing Systems, pp. 5509–5519.